TTF Natural Gas Options — User Manual

This manual covers:

- **Part 1.** Introduction — TTF options, Black-76 vs Bachelier, Greeks
- **Part 2.** `black76_ttf.py` — every function with worked examples
- **Part 3.** `ttf_market_data.py` — market data and vol surface
- **Part 4.** `ttf_hh_spread.py` — TTF/HH spread option (Margrabe 1978)
- **Part 5.** `dashboard_jupyter.ipynb` — section-by-section user guide
- **Part 6.** Financial glossary — every term used in the project

> ***Conventions used throughout the examples*** *- TTF forward:* `F = 30 EUR/MWh` *- Strikes:* `25 / 28 / 30 / 32 / 35 EUR/MWh` *- Black-76 volatility:* `sigma = 0.50` *(50% lognormal) - Bachelier volatility:* `sigma_n = 15 EUR/MWh` *(≈* `F · sigma`*) - Maturity:* `T = 0.25` *year (3 months, ACT/365) - Risk-free rate:* `r = 0.02` *(2%) - Spread (Part 4):* `F_HH = 3 USD/MMBtu`, `fx = 1.08`, `σ_TTF = 60%`, `σ_HH = 50%`, `ρ = 0.35`, `r_usd = 0.045`
>
> *Numerical values shown are rounded to 4 decimal places.*

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1. Introduction

1.1 What is a TTF option?

The **TTF** (*Title Transfer Facility*) is the Dutch virtual natural gas hub and the European benchmark. A **TTF option** grants the right (without the obligation) to buy (`call`) or sell (`put`) a TTF futures contract at a given exercise price (`strike` , in EUR/MWh) on a given expiration date.

- **Underlying**: TTF futures (one contract = physical delivery over a given month).
- **Style**: European — exercise only at expiry.
- **Quote**: EUR/MWh (1 MWh = 3.4121 MMBtu).
- **Day-count**: calendar days / 365 for the time `T` to expiry.
- **Calendar**: official **ICE TFO** (TTF Options) rule, ICE product code **TFO** (source):

> *Trading will cease when the intraday settlement price of the underlying futures contract is set, five calendar days before the start of the contract month. If that day is a non-business day or non-UK business day, expiry will occur on the nearest prior business day, except*

> *where that day is also the expiry date of the underlying futures contract, in which case expiry will occur on the preceding business day.*

The implementation uses a **UK business-day calendar** (England & Wales public holidays only).

1.2 Black-76 vs Bachelier

Two pricing models are implemented; the choice depends on the price regime.

Criterion	Black-76 (lognormal)	Bachelier (normal)
Assumption	`dF / F = sigma · dW`	`dF = sigma_n · dW`
Vol expressed in	decimal (`0.50` = 50%)	absolute EUR/MWh (`15` = 15 EUR/MWh)
Negative price allowed	**no** (F must be > 0)	**yes**
Suitable regime	normal conditions (F ≫ 0)	crisis / spreads (F low or < 0)
PCP	`C − P = e^(-rT)·(F − K)`	identical

Rule of thumb: if `F < 2 EUR/MWh` or `F` may go negative (spreads, under-balanced gas), switch to Bachelier. Otherwise, Black-76 is the standard.

At ATM (`K = F`), both models yield very similar prices when `sigma_n ≈ F · sigma`.

1.3 The Greeks

Sensitivities of the option price with respect to the input parameters.

Greek	Definition	Unit	Convention used here
Delta Δ	`∂V / ∂F`	dimensionless	discounted by `e^(-rT)`
Gamma Γ	`∂²V / ∂F²`	per EUR/MWh	identical for call/put
Vega ν	`∂V / ∂σ`	EUR/MWh per unit of vol	per 1.00 (100%) of vol
Theta Θ	`∂V / ∂t`	EUR/MWh per day	per calendar day (negative if long)
Rho ρ	`∂V / ∂r`	EUR/MWh per 1 pp	per percentage point
Vanna	`∂Δ / ∂σ`	mixed	cross derivative
Volga	`∂²V / ∂σ²`	EUR/MWh	vol convexity

Practical conventions:

- `Δ_call − Δ_put = e^(-rT)` (forward delta parity).
- `Γ`, `ν` are identical for call and put under both Black-76 and Bachelier.
- `Θ` here is divided by 365 → **per day**, not per year.
- `ρ` here is divided by 100 → per **percentage point** of rate change.

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2. `black76_ttf.py`

A single module, with no external dependency beyond `scipy.stats.norm` and `scipy.optimize.brentq`. It covers:

- The **ICE TFO** (TTF Options) expiry calendar with a UK business-day calendar
- Time-to-expiry helpers (`T` in years = calendar days / 365)
- A contract code parser (`TTFM26`, `Jun26`, …)
- **Black-76** and **Bachelier** pricing (call / put / dispatcher)
- The full set of Greeks (Δ, Γ, ν, Θ, ρ, vanna, volga) for both models
- **Implied volatility** solvers (Brent)
- **Delta** → **strike** inversion

2.1 ICE TFO expiry calendar

ICE product code: **TFO**. Source: https://www.ice.com/products/71085679/Dutch-TTF-Natural-Gas-Options-Futures-Style-Margin.

The official rule is:

> *Trading will cease when the intraday settlement price of the underlying futures contract is set, five calendar days before the start of the contract month. If that day is a non-business day or non-UK business day, expiry will occur on the nearest prior business day, except where that day is also the expiry date of the underlying futures contract, in which case expiry will occur on the preceding business day.*

The holiday calendar is **UK public holidays** (England & Wales) only: New Year's Day, Good Friday, Easter Monday, the early-May, spring and summer bank holidays, Christmas Day and Boxing Day, with the standard substitution to the next weekday when a fixed-date holiday falls on a weekend.

Computed expiries for the 12 monthly contracts of 2026:

Contract	Futures LTD	Option Expiry	Day
Jan-26	31 Dec 2025	24 Dec 2025	Wed
Feb-26	30 Jan 2026	27 Jan 2026	Tue
Mar-26	27 Feb 2026	24 Feb 2026	Tue
Apr-26	31 Mar 2026	27 Mar 2026	Fri
May-26	30 Apr 2026	24 Apr 2026	Fri
Jun-26	29 May 2026	27 May 2026	Wed
Jul-26	30 Jun 2026	26 Jun 2026	Fri
Aug-26	31 Jul 2026	27 Jul 2026	Mon
Sep-26	28 Aug 2026	27 Aug 2026	Thu
Oct-26	30 Sep 2026	25 Sep 2026	Fri
Nov-26	30 Oct 2026	27 Oct 2026	Tue
Dec-26	30 Nov 2026	26 Nov 2026	Thu

`ttf_expiry_date(contract_month: int, contract_year: int) -> date`

Official TTF option expiry date for a given delivery month.

```
from datetime import date
from black76_ttf import ttf_expiry_date

ttf_expiry_date(6, 2026)   # TTFM26: June 2026 delivery
# -> date(2026, 5, 27)

ttf_expiry_date(1, 2026)   # TTFF26: Dec-25 holiday shifts to Dec 24
# -> date(2025, 12, 24)

ttf_expiry_date(3, 2026)   # TTFH26: March 2026 delivery
# -> date(2026, 2, 24)
```

Algorithm (direct application of the ICE TFO rule):

1. Candidate = 1st of delivery month − 5 calendar days.
2. If not a UK business day → roll back to the nearest prior UK business day.
3. If that day equals the futures LTD → roll back one additional UK business day.

`ttf_time_to_expiry(contract_month, contract_year, reference=None) -> float`

Calendar days to expiry, divided by 365:

```
t = (expiry_date - reference).days / 365
```

No business-day adjustment, no day-count convention.

```
from datetime import date
from black76_ttf import ttf_time_to_expiry

ttf_time_to_expiry(6, 2026, reference=date(2026, 4, 23))
# -> 0.0932  (34 days / 365)
```

`ttf_next_expiries(n=6, reference=None) -> list[tuple[str, date]]`

The next `n` TTF expiries (ICE codes + dates), sorted ascending.

```
from datetime import date
from black76_ttf import ttf_next_expiries

ttf_next_expiries(3, reference=date(2026, 4, 23))
# -> [('TTFK26', date(2026, 4, 24)),
#     ('TTFM26', date(2026, 5, 27)),
#     ('TTFN26', date(2026, 6, 26))]
```

`ttf_is_business_day(d: date) -> bool`

UK business day — Mon–Fri excluding UK public holidays only (England & Wales): New Year's Day, Good Friday, Easter Monday, the early-May, spring and summer bank holidays, Christmas Day and Boxing Day (with the standard weekend-substitution rule).

```
from datetime import date
from black76_ttf import ttf_is_business_day

ttf_is_business_day(date(2025, 12, 25))   # Christmas
# -> False
ttf_is_business_day(date(2026, 4, 6))     # Easter Monday
# -> False
ttf_is_business_day(date(2026, 5, 25))    # Spring bank holiday
# -> False
```

2.2 Contract code parser

`t_from_contract(contract: str, reference=None) -> float`

Accepts the ICE code (`TTFH26`) or a 3-letter month abbreviation (`Mar26`, `Mar2026`) and returns `T` via `ttf_time_to_expiry` (calendar days / 365 to the ICE TFO expiry).

```
from datetime import date
from black76_ttf import t_from_contract

t_from_contract("TTFM26", reference=date(2026, 4, 1))
t_from_contract("Jun26",  reference=date(2026, 4, 1))
t_from_contract("Mar2026", reference=date(2026, 1, 15))
```

Raises `ValueError` if the format is not recognized.

2.3 Black-76 pricing

Supported ICE month codes: `F G H J K M N Q U V X Z`.

`b76_call(F, K, T, r, sigma) -> float`

```
from black76_ttf import b76_call

b76_call(F=30, K=30, T=0.25, r=0.02, sigma=0.50)
# -> 2.9670   EUR/MWh   (ATM call, 3 months)

b76_call(F=30, K=25, T=0.25, r=0.02, sigma=0.50)
# -> 6.4124   EUR/MWh   (ITM call)

b76_call(F=30, K=35, T=0.25, r=0.02, sigma=0.50)
# -> 1.2197   EUR/MWh   (OTM call)
```

`b76_put(F, K, T, r, sigma) -> float`

```
from black76_ttf import b76_put

b76_put(F=30, K=30, T=0.25, r=0.02, sigma=0.50)
# -> 2.9670   EUR/MWh   (ATM put, 3 months)

b76_put(F=30, K=35, T=0.25, r=0.02, sigma=0.50)
# -> 6.1448   EUR/MWh   (ITM put)
```

> ***PCP check**: for `K = 35`, `C − P = 1.2197 − 6.1448 = −4.9251` while `e^(-0.02·0.25)·(30 − 35) = −4.9750`. Exact difference → PCP holds.*

`b76_price(F, K, T, r, sigma, option_type='call') -> float`

Generic dispatcher:

```
from black76_ttf import b76_price

b76_price(30, 30, 0.25, 0.02, 0.50, "call")   # -> 2.9670
b76_price(30, 30, 0.25, 0.02, 0.50, "put")    # -> 2.9670
```

Raises `ValueError` if `option_type` $\notin$ `{'call', 'put'}`.

`b76_price_ttf(F, K, contract, r, sigma, option_type='call', reference=None) -> float`

Variant where `T` is derived from a contract code.

```
from datetime import date
from black76_ttf import b76_price_ttf

b76_price_ttf(F=30, K=30, contract="TTFM26",
              r=0.02, sigma=0.50,
              option_type="call",
              reference=date(2026, 4, 1))
# -> 2.2530   EUR/MWh   (T = 52 / 365 = 0.1425)
```

2.4 Bachelier pricing

For forwards close to zero or negative, or for spreads.

`bach_call(F, K, T, r, sigma_n) -> float`

```
from black76_ttf import bach_call

bach_call(F=30, K=30, T=0.25, r=0.02, sigma_n=15)
# -> 2.9770   EUR/MWh

bach_call(F=-3, K=0, T=60/365, r=0.02, sigma_n=6)
# -> 0.5860   EUR/MWh   (negative forward, crisis scenario)
```

`bach_put(F, K, T, r, sigma_n) -> float`

```
from black76_ttf import bach_put

bach_put(F=30, K=30, T=0.25, r=0.02, sigma_n=15)
# -> 2.9770
```

`bach_price(F, K, T, r, sigma_n, option_type='call') -> float`

Dispatcher.

```
from black76_ttf import bach_price

bach_price(30, 35, 0.25, 0.02, 15, "put")
# -> 7.6770
```

`bach_price_ttf(F, K, contract, r, sigma_n, option_type='call', reference=None) -> float`

Same as `b76_price_ttf` but using Bachelier.

```
from datetime import date
from black76_ttf import bach_price_ttf

bach_price_ttf(F=30, K=30, contract="Jun26",
               r=0.02, sigma_n=15,
               option_type="call",
               reference=date(2026, 4, 1))
# -> 2.2519
```

2.5 Black-76 Greeks

All the functions below take `(F, K, T, r, sigma, option_type='call')` except `b76_gamma`, `b76_vega`, `b76_vanna`, `b76_volga` which are identical for call and put (no `option_type` argument).

```
from black76_ttf import (
    b76_delta, b76_gamma, b76_vega, b76_theta, b76_rho,
    b76_vanna, b76_volga, b76_greeks,
)

F, K, T, r, sigma = 30, 30, 0.25, 0.02, 0.50

b76_delta(F, K, T, r, sigma, "call")   # -> +0.5470
b76_delta(F, K, T, r, sigma, "put")    # -> -0.4480
b76_gamma(F, K, T, r, sigma)           # -> +0.0525
b76_vega (F, K, T, r, sigma)           # -> +5.9069   (per 1.00 of vol)
b76_theta(F, K, T, r, sigma, "call")   # -> -0.0245   (per day)
b76_rho  (F, K, T, r, sigma, "call")   # -> -0.0074   (per 1 pp)
b76_vanna(F, K, T, r, sigma)           # -> +0.0984
b76_volga(F, K, T, r, sigma)           # -> -0.1846
```

`b76_greeks(F, K, T, r, sigma, option_type='call') -> B76Greeks`

All the Greeks in a single call (`B76Greeks` dataclass with fields `delta`, `gamma`, `vega`, `theta`, `rho`, `vanna`, `volga`).

```
g = b76_greeks(F=30, K=30, T=0.25, r=0.02, sigma=0.50, option_type="call")
g.delta   # +0.5470
g.gamma   # +0.0525
g.vega    # +5.9069
g.theta   # -0.0245
g.rho     # -0.0074
g.vanna   # +0.0984
g.volga   # -0.1846
```

> ***Vega convention**: per unit of decimal vol (1.00 = 100%). For the "per 1 vol point" sensitivity (1% = 0.01), divide vega by 100.*
>
> ***Rho convention**: already divided by 100, so directly in EUR/MWh per percentage point of rate change.*

2.6 Bachelier Greeks

Same signatures, with `sigma_n` instead of `sigma`.

```
from black76_ttf import (
    bach_delta, bach_gamma, bach_vega, bach_theta, bach_rho,
    bach_vanna, bach_volga, bach_greeks,
)

F, K, T, r, sigma_n = 30, 30, 0.25, 0.02, 15

bach_delta(F, K, T, r, sigma_n, "call")   # -> +0.4975
bach_delta(F, K, T, r, sigma_n, "put")    # -> -0.4975
bach_gamma(F, K, T, r, sigma_n)           # -> +0.0530
bach_vega (F, K, T, r, sigma_n)           # -> +0.1985
bach_theta(F, K, T, r, sigma_n, "call")   # -> -0.0246
bach_rho  (F, K, T, r, sigma_n, "call")   # -> -0.0074
bach_vanna(F, K, T, r, sigma_n)           # 0.0
bach_volga(F, K, T, r, sigma_n)           # -0.0132
```

> *At ATM under Bachelier, `vanna = 0` exactly (by symmetry in `d = (F-K)/(σ√T)`).*

`bach_greeks(...) -> BachGreeks`

```
g = bach_greeks(30, 30, 0.25, 0.02, 15, "call")
g.delta, g.gamma, g.vega, g.theta, g.rho, g.vanna, g.volga
# (0.4975, 0.0530, 0.1985, -0.0246, -0.0074, 0.0, -0.0132)
```

2.7 Implied volatility solvers

Brent method, `xtol = 1e-8`, max 300 iterations.

`b76_implied_vol(market_price, F, K, T, r, option_type='call', sigma_lo=1e-6, sigma_hi=20.0) -> float`

Round-trip price → IV → sigma:

```
from black76_ttf import b76_call, b76_implied_vol

p = b76_call(30, 30, 0.25, 0.02, 0.50)   # 2.9670
b76_implied_vol(p, F=30, K=30, T=0.25, r=0.02, option_type="call")
# -> 0.5000
```

Raises `ValueError` if `market_price` falls outside the `[intrinsic, sigma_hi]` corridor.

`bach_implied_vol(market_price, F, K, T, r, option_type='call', sigma_lo=1e-6, sigma_hi=500.0) -> float`

Inverse for the normal vol in EUR/MWh.

```
from black76_ttf import bach_call, bach_implied_vol

p = bach_call(30, 30, 0.25, 0.02, 15)   # 2.9770
bach_implied_vol(p, F=30, K=30, T=0.25, r=0.02, option_type="call")
# -> 15.0000
```

2.8 Delta → strike inversion

Useful for converting a 25Δ / 50Δ / 75Δ quote into a strike.

`b76_delta_to_strike(delta_target, F, T, r, sigma, option_type='call', K_lo=None, K_hi=None) -> float`

```
from black76_ttf import b76_delta_to_strike

# Strike of the 25-delta call (typically OTM)
b76_delta_to_strike(delta_target=0.25,
                    F=30, T=0.25, r=0.02, sigma=0.50,
                    option_type="call")
# -> 35.0826   EUR/MWh

# Strike of the 25-delta put (the target delta is negative for a put)
b76_delta_to_strike(delta_target=-0.25,
                    F=30, T=0.25, r=0.02, sigma=0.50,
                    option_type="put")
# -> 25.5749   EUR/MWh
```

Defaults for `K_lo` / `K_hi`: `[F · 0.01, F · 10]`. Raises `ValueError` if the target is not reachable within the bracket.

`bach_delta_to_strike(delta_target, F, T, r, sigma_n, option_type='call', K_lo=None, K_hi=None) -> float`

Bachelier variant — useful for negative or very low forwards.

```
from black76_ttf import bach_delta_to_strike

bach_delta_to_strike(delta_target=0.25,
                     F=30, T=0.25, r=0.02, sigma_n=15,
                     option_type="call")
# -> 35.0570   EUR/MWh
```

Defaults for `K_lo` / `K_hi`: `[F − 10·σ_n·√T, F + 10·σ_n·√T]`.

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3. `ttf_market_data.py`

End-to-end market data layer for the TTF natural gas curve and its volatility surface. The module is **fully self-contained** (no `yfinance` dependency) and gracefully falls back to a synthetic curve when the public quote feed is unreachable.

What is covered:

- TTF expiry calendar (ICE TFO official rule, UK business-day calendar)
- Forward curve fetching (Yahoo Finance via `requests`) with synthetic fallback
- **Manual / CSV forward-curve loader** (§ 3.10) with a bundled Jun-26 → Dec-27 sample

- Volatility surface construction (strikes × maturities, parametric smile)
- **Delta-quoted vol-surface utilities** (§ 3.10): `update_vol_surface`, `display_vol_surface`, plus a TTF-style downside-skew sample
- SABR market calibration (Hagan et al. 2002)
- CSV / JSON export for downstream pricing modules

> *The new manual helpers in § 3.10 import `ttf_expiry_date` and `ttf_time_to_expiry` directly from `black76_ttf` so all dates and time-to-expiry values stay aligned with the official ICE TFO rule.*
>
> ***External dependencies*** *: `numpy`, `pandas`, `scipy`, `requests`. Make sure `requirements.txt` lists `pandas>=2.0` and `requests>=2.31` in addition to `numpy` and `scipy`.*

3.1 `TTFContract` dataclass

A single monthly contract description.

Field	Type	Meaning
`delivery_month`	`int`	1–12
`delivery_year`	`int`	e.g. `2026`
`expiry_date`	`date`	ICE TFO option expiry
`futures_expiry_date`	`date`	Underlying TTF futures last trading day
`contract_code`	`str`	ICE-style code, e.g. `"TTFH26"`
`T`	`float`	Calendar days to option expiry / 365

3.2 `TTFExpiryCalendar`

Convenience class wrapping the official **ICE TFO** rule (product code **TFO**) so it can be used with a fixed reference date. All expiry dates delegate to `ttf_expiry_date` from `black76_ttf`, so the calendar always matches the spec:

> *Trading will cease when the intraday settlement price of the underlying futures contract is set, five calendar days before the start of the contract month. If that day is a non-business day or non-UK business day, expiry will occur on the nearest prior business day, except where that day is also the expiry date of the underlying futures contract, in which case expiry will occur on the preceding business day.*

Source: https://www.ice.com/products/71085679/Dutch-TTF-Natural-Gas-Options-Futures-Style-Margin. The holiday calendar is **UK public holidays** (England & Wales) only.

```
from datetime import date
from ttf_market_data import TTFExpiryCalendar

cal = TTFExpiryCalendar(reference_date=date(2026, 4, 20))
```

`futures_expiry_date(delivery_year, delivery_month) -> date`

Last UK business day of the month immediately before the contract month.

```
cal.futures_expiry_date(2026, 6)   # TTFM26
# -> date(2026, 5, 29)             (Friday)
```

`expiry_date(delivery_year, delivery_month) -> date`

ICE TFO option expiry. Delegates directly to `black76_ttf.ttf_expiry_date(delivery_month, delivery_year)`.

```
cal.expiry_date(2026, 6)
# -> date(2026, 5, 27)             (Wed)
```

`contract_code(delivery_year, delivery_month) -> str`

Returns the ICE-style code (`TTF` + month code letter + last two digits of the year).

```
cal.contract_code(2026, 3)         # -> "TTFH26"
cal.contract_code(2027, 12)        # -> "TTFZ27"
```

`time_to_expiry(expiry) -> float`

Time `T` = `(expiry - reference).days / 365` (calendar days only). Returns `0` if the expiry is already in the past.

```
cal.time_to_expiry(date(2026, 5, 27))
# -> 0.1014   (37 days / 365)
```

`active_contracts(n=12) -> list[TTFContract]`

The next `n` monthly TTF contracts (sorted ascending), skipping any whose expiry has already passed relative to the reference date.

```
contracts = cal.active_contracts(n=3)
for c in contracts:
    print(c.contract_code, c.expiry_date, round(c.T, 4))
# TTFK26  2026-04-24  0.0110
# TTFM26  2026-05-27  0.0932
# TTFN26  2026-06-26  0.1753
```

`expiry_for_tenor(tenor_years) -> date`

Returns the expiry date of the contract closest to a given tenor expressed in years. Useful when interpolating a vol smile by tenor rather than by contract.

```
cal.expiry_for_tenor(0.50)
# -> date(2026, 10, 23)   (≈ 6 months from 2026-04-20)
```

3.3 `ForwardPoint` dataclass

A single (contract, price) observation on the curve.

Field	Type	Meaning
`contract_code`	`str`	e.g. `"TTFM26"`
`delivery_month`	`int`	1–12
`delivery_year`	`int`	e.g. `2026`
`expiry_date`	`date`	Options expiry
`T`	`float`	Time to expiry (years, ACT/365)
`forward_price`	`float`	EUR/MWh, rounded to 4 decimals
`source`	`str`	`"yahoo_finance"` or `"synthetic"`

3.4 `TTFForwardCurve`

Fetches and interpolates the TTF forward curve.

```
from datetime import date
from ttf_market_data import TTFForwardCurve

curve = TTFForwardCurve(
    reference_date=date(2026, 4, 20),
    risk_free_rate=0.03,
    timeout=10,
).load()
```

The constructor takes:

- `reference_date` : valuation date (defaults to `date.today()`)
- `risk_free_rate` : EUR risk-free rate used for cost-of-carry projection
- `timeout` : HTTP timeout in seconds for the Yahoo Finance request

`.load()` fetches the front-month TTF spot from Yahoo Finance (`ticker = "TTF=F"`); on failure (network error, throttling, schema change), it logs a warning and falls back to a representative synthetic spot of `35.0 EUR/MWh`. Each forward is then projected with a small seasonal premium:

```
seasonal(m) = 0.03 · sin(2π · (m − 1) / 12)
F(c) = spot · exp((r + seasonal(m)) · T)
```

`forward_price(T) -> float`

Linearly interpolated forward price for a tenor `T` (years). Flat extrapolation outside the loaded range.

```
curve.forward_price(0.25)   # -> 35.40 EUR/MWh   (illustrative)
curve.forward_price(1.00)   # -> 36.85 EUR/MWh
```

`to_dataframe() -> pandas.DataFrame`

Returns the full curve as a `DataFrame` with columns `contract_code, delivery_month, delivery_year, expiry_date, T, forward_price, source`.

```
curve.to_dataframe().head()
#   contract_code  delivery_month  delivery_year  expiry_date     T
forward_price       source
# 0       TTFK26               5           2026   2026-04-24  0.014
35.0145  yahoo_finance
# 1       TTFM26               6           2026   2026-05-22  0.090
35.1842  yahoo_finance
# ...
```

> ***Tip*** *: if Yahoo Finance is unavailable in your network (corporate proxy, sandboxed CI), wrap the load in a* `try/except` *— the module already logs a warning but still returns a usable synthetic curve.*

3.5 `VolSmile` and `VolatilitySurface`

```
@dataclass
class VolSmile:
    T: float                  # time to expiry (years)
    contract_code: str        # e.g. 'May26'
    F: float                  # forward at this tenor
    strikes: list[float]      # 9 strikes by default
    vols: list[float]         # Black-76 lognormal vols
    model: str = "black76"    # "black76" or "bachelier"
```

`VolatilitySurface` is a container of smiles indexed by maturity. It provides:

- `add_smile(smile)` — append a smile (and invalidate the bilinear cache)
- `vol(K, T) -> float` — bilinear interpolation in `(strike, maturity)` space
- `to_dataframe() -> DataFrame` — long format (one row per `(T, K)`)

```
v = surface.vol(K=32.0, T=0.50)        # interpolated lognormal vol
df = surface.to_dataframe()             # for plotting / export
```

> *The interpolator is built on first call to `.vol(...)` and re-cached automatically when a new smile is appended.*

3.6 `VolatilitySurfaceBuilder`

Builds a parametric TTF vol surface when no live option quotes are available.

```
from ttf_market_data import VolatilitySurfaceBuilder

builder = VolatilitySurfaceBuilder(
    forward_curve=curve,
    reference_date=date(2026, 4, 20),
    n_strikes=9,
)
surface = builder.build()
```

Inputs (all optional with sensible defaults):

Argument	Default	Meaning
`atm_vols`	`{1/12: 0.65, 2/12: 0.58, 3/12: 0.52, 6/12: 0.46, 9/12: 0.42, 1.0: 0.40, 2.0: 0.38}`	ATM lognormal vol per tenor
`rr25`	`-0.03` per tenor	25-delta risk-reversal (negative ⇒ put vol > call vol)
`bf25`	`+0.015` per tenor	25-delta butterfly (smile convexity)

For each tenor, the builder converts (ATM, RR25, BF25) into 25Δ call and 25Δ put vols, inverts those to strikes via `_delta_to_strike`, and interpolates the vols along the resulting strike grid in log-moneyness space:

```
σ_25c = ATM + 0.5 · RR25 + BF25
σ_25p = ATM − 0.5 · RR25 + BF25
```

If the forward at a given tenor falls below `2 EUR/MWh`, the smile is tagged `model="bachelier"` so downstream code knows to switch pricing models.

3.7 `SABRParams` and `MarketCalibration`

```
@dataclass
class SABRParams:
    alpha: float   # initial vol level
    beta: float    # CEV exponent (0.5 — typical for energy markets)
    rho: float     # vol-spot correlation
    nu: float      # vol-of-vol
```

`MarketCalibration` runs SABR (Hagan et al. 2002) on every Black-76 smile in the surface; Bachelier smiles are skipped (with a warning).

```
from ttf_market_data import MarketCalibration

calib = MarketCalibration(surface).calibrate_all()
calib.to_dataframe()
#       T   alpha  beta      rho      nu
# 0  0.083  0.471  0.50  -0.214  0.521
# 1  0.167  0.434  0.50  -0.198  0.488
# ...
```

Beta is fixed at `0.5`. The optimiser is `scipy.optimize.minimize` with L-BFGS-B and bounds `alpha ∈ [1e-4, 5]`, `rho ∈ [−0.999, 0.999]`, `nu ∈ [1e-4, 5]`.

3.8 Export utilities

Three helpers serialise to both CSV and JSON. `path` is the **prefix without extension**; the helpers append `.csv` / `.json`.

`export_forward_curve(curve, path)`

Writes `path.csv` and `path.json`. The JSON wraps the records under a top-level `{ "reference_date": "...", "curve": [...] }` envelope.

```
from ttf_market_data import export_forward_curve

export_forward_curve(curve, "ttf_output/ttf_forward_curve")
# -> ttf_output/ttf_forward_curve.csv
# -> ttf_output/ttf_forward_curve.json
```

`export_vol_surface(surface, path)`

Writes three files: `path.csv` (long format), `path_pivot.csv` (wide pivot, rows = `T`, columns = strike), and `path.json` (records under `{ "reference_date": "...", "surface": [...] }`).

`export_sabr_params(calibration, path)`

Writes `path.csv` and `path.json` with one row per calibrated tenor.

`export_all(output_dir=".", reference_date=None, risk_free_rate=0.03)`

Runs the full pipeline (forward curve → vol surface → SABR calibration) and writes everything to `output_dir`. Returns a `dict` with the three DataFrames for in-process use.

```
from ttf_market_data import export_all
from datetime import date

dfs = export_all(output_dir="ttf_output", reference_date=date(2026, 4, 20))
sorted(dfs.keys())
# ['forward_curve', 'sabr_params', 'vol_surface']
```

This is the recommended one-shot entry point for the rest of the project — the dashboard, the spread pricer and the structures module all consume the JSON files written here.

3.9 End-to-end example

```
from datetime import date
from ttf_market_data import (
    TTFExpiryCalendar, TTFForwardCurve,
    VolatilitySurfaceBuilder, MarketCalibration,
)

ref = date(2026, 4, 20)

# 1. Calendar — next 6 active TTF contracts
cal = TTFExpiryCalendar(ref)
for c in cal.active_contracts(n=6):
    print(f"{c.contract_code}  expiry={c.expiry_date}  T={c.T:.4f}y")

# 2. Forward curve (Yahoo Finance, with synthetic fallback)
curve = TTFForwardCurve(ref, risk_free_rate=0.03).load()
print(curve.to_dataframe()[["contract_code", "T", "forward_price", "source"]])

# 3. Vol surface (parametric, ATM term-structure + 25Δ smile)
surface = VolatilitySurfaceBuilder(curve, ref).build()
print(surface.vol(K=32.0, T=0.25))   # interpolated lognormal vol

# 4. SABR calibration (per tenor)
calib = MarketCalibration(surface).calibrate_all()
print(calib.to_dataframe())
```

3.10 Manual / CSV forward curve and vol-surface helpers

A small set of helpers for analytics notebooks that prefer hand-typed or CSV market data rather than a live feed. They all anchor on the official **ICE TFO** expiry calendar — `ttf_expiry_date` and `ttf_time_to_expiry` are imported from `black76_ttf` and never recomputed locally.

`load_ttf_forward_curve(source='manual', data=None, filepath=None, reference=None) -> pd.DataFrame`

Two input modes:

`source`	What it does
`'manual'`	Uses the `data` dict `{delivery_month: forward_price}`. If `data` is `None`, falls back to the bundled `SAMPLE_TTF_FORWARD_CURVE`.
`'csv'`	Reads `filepath` (columns `delivery_month`, `forward_price`).

Output DataFrame columns:

Column	Source
`delivery_month`	input label, e.g. `"Jun-26"`
`expiry_date`	`ttf_expiry_date(month, year)`
`time_to_expiry`	`ttf_time_to_expiry(month, year, reference)`
`forward_price`	input price (EUR/MWh)

```
from datetime import date
from ttf_market_data import load_ttf_forward_curve

# 1. Manual — pass a {delivery_month: forward_price} dict
fc = load_ttf_forward_curve(
    source="manual",
    data={"Jun-26": 30.5, "Jul-26": 31.2, "Aug-26": 32.0},
    reference=date(2026, 4, 23),
)

# 2. CSV — file with columns (delivery_month, forward_price)
fc = load_ttf_forward_curve(source="csv", filepath="ttf_forwards.csv")

# 3. No arguments → bundled sample (Jun-26 → Dec-27, 19 contracts)
fc = load_ttf_forward_curve(reference=date(2026, 4, 23))
```

`SAMPLE_TTF_FORWARD_CURVE`

Bundled 19-contract sample, EUR/MWh, with a realistic seasonal shape (winter peaks ≈ 35, summer troughs ≈ 28-29). Useful for demos and tests:

Contract	F (EUR/MWh)	Contract	F (EUR/MWh)	Contract	F (EUR/MWh)
Jun-26	28.50	Jan-27	35.00	Aug-27	30.50
Jul-26	29.00	Feb-27	34.20	Sep-27	31.80
Aug-26	30.20	Mar-27	32.50	Oct-27	33.50
Sep-26	31.50	Apr-27	30.80	Nov-27	34.50
Oct-26	33.20	May-27	29.50	Dec-27	35.00
Nov-26	34.50	Jun-27	29.00		
Dec-26	35.20	Jul-27	29.50		

`update_vol_surface(forward_curve, vol_surface) -> pd.DataFrame`

Joins a delta-quoted vol surface with the forward curve. The input `vol_surface` is a nested dict `{delivery_month: {call_delta_pillar: vol}}` — the same shape as `SAMPLE_VOL_SURFACE` below.

For each `(delivery_month, delta_pillar, vol)` triple:

- **Strike** — at the ATM pillar (`Δ = 0.50`) the delta-neutral rule:

  $K_atm = F \cdot \exp(-\sigma^2 T / 2)$

At any other pillar, the Black-76 inversion of the call delta:

```
K = F · exp(0.5 σ² T − Φ⁻¹(δ) · σ √T)
```

- **Delta** — re-computed under Black-76 forward delta `N(d₁)` at that `(K, σ)` so each row is internally consistent. Note that at the ATM-DN strike the realised model delta is `N(σ √T)`, which drifts above 0.50 for longer maturities — this is expected.
- **Moneyness** — `K / F`.

Output columns:

```
delivery_month, expiry_date, time_to_expiry, forward_price,
strike, implied_vol, delta, moneyness
```

(plus a `delta_pillar` column carrying the input pillar so that `display_vol_surface` can pivot the smile axis cleanly).

```
from ttf_market_data import update_vol_surface, SAMPLE_VOL_SURFACE
vs = update_vol_surface(fc, SAMPLE_VOL_SURFACE)
print(vs.head(7).to_string(index=False))
```

`SAMPLE_VOL_SURFACE`

A TTF-style downside-skew surface aligned 1-for-1 with `SAMPLE_TTF_FORWARD_CURVE`:

- ATM ≈ **50%** at the front month → ≈ **37%** around 1.5 years
- 10-Δ-call to 90-Δ-call wing spread of about **14 vol points** (puts richer than calls — the standard TTF / energy skew)
- Five pillars per smile: `Δ ∈ {0.10, 0.25, 0.50, 0.75, 0.90}`

`display_vol_surface(vol_surface_df) -> pd.DataFrame`

Pretty-prints the surface as a `maturity × call-delta-pillar` grid:

- Rows ordered by ascending `time_to_expiry`

- Columns labelled `Δ=0.10 … Δ=0.90`, with the ATM column suffixed `(ATM)` so it stands out at a glance
- Vols formatted as percentages with one decimal place
- The function prints the grid to `stdout` and returns the formatted pivot DataFrame for further export

```
from ttf_market_data import display_vol_surface
display_vol_surface(vs)
```

```
TTF vol surface — rows: maturity, cols: call-delta pillar
ATM column is the delta-neutral strike (K = F·exp(−σ²T/2))

          Δ=0.10  Δ=0.25 Δ=0.50 (ATM)  Δ=0.75  Δ=0.90
Delivery
Jun-26     43.0%   47.0%        50.0%   53.0%   57.0%
Jul-26     42.0%   46.0%        49.0%   52.0%   56.0%
Aug-26     41.0%   45.0%        48.0%   51.0%   55.0%
…
Nov-27     31.0%   35.0%        38.0%   41.0%   45.0%
Dec-27     30.0%   34.0%        37.0%   40.0%   44.0%
```

End-to-end manual example

```
from datetime import date
from ttf_market_data import (
    load_ttf_forward_curve,
    update_vol_surface,
    display_vol_surface,
    SAMPLE_VOL_SURFACE,
)

ref = date(2026, 4, 23)

fc = load_ttf_forward_curve(reference=ref)        # bundled sample
vs = update_vol_surface(fc, SAMPLE_VOL_SURFACE)
display_vol_surface(vs)
```

---

4. `ttf_hh_spread.py`

A pricer for **exchange options** on the TTF − HH basis using **Margrabe's formula** (1978). The module covers:

- Unit conversion EUR/MWh ⇄ USD/MMBtu (`1 MWh = 3.412142 MMBtu`)
- Margrabe call/put on `(F_TTF, F_HH)` in USD/MMBtu
- All first-order Greeks plus `vega_ρ` (sensitivity to correlation)

- Implied correlation solver (Brent, monotone in ρ)
- Correlation- and vol-sensitivity tables
- A formatted summary printer

> ***Conventions for this section*** *- TTF forward:* `F_ttf_eur = 30 EUR/MWh` *- Henry Hub forward:* `F_hh = 3 USD/MMBtu` *- EUR/USD:* `fx_eurusd = 1.08` *- Vols:* `σ_TTF = 0.60` *(60%),* `σ_HH = 0.50` *(50%) - Correlation:* `ρ = 0.35` *- USD risk-free:* `r_usd = 0.045`

4.1 Background — exchange options and Margrabe

For two log-normal forwards `F_TTF`, `F_HH` (both in **USD/MMBtu**) with vols `σ_TTF`, `σ_HH` and instantaneous correlation `ρ`, the **spread vol** is:

```
σ_s = √(σ_TTF² + σ_HH² − 2·ρ·σ_TTF·σ_HH)
```

The price of the exchange call (Margrabe, 1978) is:

```
C = e^(-rT) · [F_TTF · N(d₁) − F_HH · N(d₂)]
P = e^(-rT) · [F_HH · N(−d₂) − F_TTF · N(−d₁)]

with  d₁ = [ln(F_TTF / F_HH) + ½·σ_s²·T] / (σ_s·√T)
      d₂ = d₁ − σ_s·√T
```

Intuition:

- **ρ → +1** (assets co-move) → `σ_s ↓` → option **cheaper**
- **ρ → −1** (assets diverge) → `σ_s ↑` → option **more expensive**
- The implied correlation backed out of a market quote is the market's view of TTF/HH co-movement.

Typical LNG-driven regime (2022–2026):

Quantity	Range
TTF forward	25 – 45 EUR/MWh (≈ 8 – 14 USD/MMBtu after FX)
Henry Hub forward	2 – 5 USD/MMBtu
Spread TTF − HH	5 – 12 USD/MMBtu (LNG netback drives convergence)
Implied correlation	≈ 0.20 – 0.55

4.2 Constants and unit conversions

`MWH_TO_MMBTU = 3.412142`

Exact energy equivalence factor: `1 MWh = 3.412142 MMBtu`.

`ttf_eur_to_usd(F_ttf_eur, fx_eurusd) -> float`

Converts a TTF forward from EUR/MWh to USD/MMBtu: `F_usd = F_eur · fx_eurusd / 3.412142`.

```
from ttf_hh_spread import ttf_eur_to_usd

ttf_eur_to_usd(F_ttf_eur=30.0, fx_eurusd=1.08)
# -> 9.4955   USD/MMBtu
```

`ttf_usd_to_eur(F_ttf_usd, fx_eurusd) -> float`

Inverse conversion.

```
from ttf_hh_spread import ttf_usd_to_eur

ttf_usd_to_eur(F_ttf_usd=10.0, fx_eurusd=1.08)
# -> 31.5939   EUR/MWh
```

`spread_usd_to_eur(spread_usd, fx_eurusd) -> float`

Converts any USD/MMBtu spread or option premium back into EUR/MWh.

```
from ttf_hh_spread import spread_usd_to_eur

spread_usd_to_eur(spread_usd=1.0, fx_eurusd=1.08)
# -> 3.1594    EUR/MWh
```

4.3 Margrabe pricing — core

The functions in this section work in **USD/MMBtu** for both forwards. Use them when the TTF forward has already been converted (or for unit testing).

`margrabe_price(F_ttf, F_hh, T, r, sigma_ttf, sigma_hh, rho, option_type='call') -> float`

```
from ttf_hh_spread import margrabe_price

# ATM-like spread: F_TTF ≈ F_HH (USD/MMBtu)
margrabe_price(F_ttf=9.50, F_hh=9.00, T=0.25, r=0.045,
               sigma_ttf=0.60, sigma_hh=0.50, rho=0.35,
               option_type="call")
# -> 1.4129   USD/MMBtu

margrabe_price(9.50, 9.00, 0.25, 0.045, 0.60, 0.50, 0.35, "put")
# -> 0.9185   USD/MMBtu
```

> ***Edge cases***: *when* `T ≤ 0` *or* `σ_s < 1e-12`, *the function returns the discounted intrinsic* `e^(-rT)·max(F_TTF − F_HH, 0)` *(call) or its symmetric counterpart (put).*

Raises `ValueError` if `option_type` ∉ `{'call', 'put'}`.

`margrabe_greeks(F_ttf, F_hh, T, r, sigma_ttf, sigma_hh, rho, option_type='call') -> SpreadGreeks`

All first-order sensitivities returned in a single `SpreadGreeks` dataclass:

Field	Definition	Unit
`delta_ttf`	`∂Price/∂F_TTF`	dimensionless (∈ [0,1] for call)
`delta_hh`	`∂Price/∂F_HH`	dimensionless (∈ [−1,0] for call)
`gamma_ttf`	`∂²Price/∂F_TTF²`	per (USD/MMBtu)
`vega_ttf`	`∂Price/∂σ_TTF`	USD/MMBtu per 1.00 of vol
`vega_hh`	`∂Price/∂σ_HH`	USD/MMBtu per 1.00 of vol
`vega_rho`	`∂Price/∂ρ`	USD/MMBtu per unit correlation
`theta`	1-day finite-difference decay	USD/MMBtu per calendar day

```
from ttf_hh_spread import margrabe_greeks

g = margrabe_greeks(F_ttf=9.50, F_hh=9.00, T=0.25, r=0.045,
                    sigma_ttf=0.60, sigma_hh=0.50, rho=0.35,
                    option_type="call")
g.delta_ttf   # +0.6219
g.delta_hh    # -0.4995
g.gamma_ttf   # +0.1244
g.vega_ttf    # +1.1928   (per 1.00 of vol)
g.vega_hh     # +0.8139
g.vega_rho    # -0.8420   (negative: ↑ρ ⇒ ↓price)
g.theta       # -0.0060   (per calendar day)
```

> ***Vega convention**: per unit of decimal vol (1.00 = 100%). Divide by 100 for the "per 1 vol point" market convention.*
>
> ***`vega_rho` sign**: a higher `ρ` reduces `σ_s`, hence reduces the option premium. `vega_rho` is therefore **negative** for both call and put.*
>
> ***Theta**: computed by 1-day finite difference (`T → T − 1/365`), so already expressed per **calendar day**. Negative for a long position with positive time value.*

4.4 Full pricer (TTF in EUR/MWh)

End-to-end entry point. It accepts TTF in **EUR/MWh**, performs the FX and energy-unit conversion internally, and returns a `SpreadResult` carrying the prices in **both** USD/MMBtu and EUR/MWh, plus all the Greeks.

`spread_price(F_ttf_eur, F_hh, fx_eurusd, T, r_usd, sigma_ttf, sigma_hh, rho, option_type='call', reference=None) -> SpreadResult`

```
from ttf_hh_spread import spread_price

res = spread_price(
    F_ttf_eur=30.0, F_hh=3.0, fx_eurusd=1.08,
    T=0.25, r_usd=0.045,
    sigma_ttf=0.60, sigma_hh=0.50, rho=0.35,
    option_type="call",
)

res.F_ttf_usd      # 9.4955
res.sigma_spread   # 0.6325   (= 63.25%)
res.price          # 6.4229   USD/MMBtu
res.price_eur      # 20.2924  EUR/MWh
res.greeks.delta_ttf
```

The `T` argument may also be a **TTF contract code** (`"TTFM26"` , `"Jun26"` , `"Mar2026"`); it is then resolved via `t_from_contract` from `black76_ttf.py` .

```
from datetime import date
from ttf_hh_spread import spread_price

res = spread_price(
    F_ttf_eur=30.0, F_hh=3.0, fx_eurusd=1.08,
    T="TTFM26", r_usd=0.045,
    sigma_ttf=0.60, sigma_hh=0.50, rho=0.35,
    option_type="call",
    reference=date(2026, 4, 1),
)
# T          -> 0.1425   (52 days / 365)
# price      -> 6.4540   USD/MMBtu
# price_eur  -> 20.3907  EUR/MWh
```

The `SpreadResult` dataclass also carries the normalised inputs (`F_ttf_eur` , `F_ttf_usd` , `F_hh` , `fx_eurusd` , `sigma_ttf` , `sigma_hh` , `rho` , `sigma_spread` , `T` , `r_usd` , `option_type`) for later inspection.

4.5 Implied correlation

Brent solver, `xtol = 1e-8` , max 300 iterations. The premium is **monotone-decreasing in ρ**, so the bracket always converges if the market price lies inside the achievable corridor.

`implied_correlation(market_price, F_ttf, F_hh, T, r, sigma_ttf, sigma_hh, option_type='call', rho_lo=-0.9999, rho_hi=0.9999) -> float`

```
from ttf_hh_spread import margrabe_price, implied_correlation

# Price an ATM spread at ρ = 0.5, then back out ρ from the price
mp = margrabe_price(9.50, 9.50, 0.25, 0.045, 0.60, 0.50, 0.50, "call")
# -> 1.0399

implied_correlation(mp,
                    F_ttf=9.50, F_hh=9.50,
                    T=0.25, r=0.045,
                    sigma_ttf=0.60, sigma_hh=0.50,
                    option_type="call")
# -> 0.5000
```

Raises `ValueError` if `market_price` lies outside the achievable corridor `[price@ρ=rho_hi, price@ρ=rho_lo]` . The error message reports the bounds — typically a quote below this corridor signals a price below intrinsic.

4.6 Sensitivity helpers

`rho_sensitivity(F_ttf, F_hh, T, r, sigma_ttf, sigma_hh, option_type='call', rhos=None) -> list[(rho, price)]`

Default ρ-grid: `[-0.9, -0.7, -0.5, -0.3, -0.1, 0.0, 0.1, 0.3, 0.5, 0.7, 0.9]`.

```
from ttf_hh_spread import rho_sensitivity

rho_sensitivity(F_ttf=9.50, F_hh=9.00, T=0.25, r=0.045,
                sigma_ttf=0.60, sigma_hh=0.50,
                option_type="call",
                rhos=[-0.5, 0.0, 0.35, 0.7])
# -> [(-0.5, 1.9821), (0.0, 1.6765), (0.35, 1.4129), (0.7, 1.0651)]
```

The price is **monotone-decreasing in ρ**, as expected for an exchange option.

`vol_sensitivity(F_ttf, F_hh, T, r, sigma_ttf, sigma_hh, rho, option_type='call') -> dict`

One-way ±5 vol-point bumps for each of `σ_TTF`, `σ_HH`, plus a ±0.10 bump on ρ — useful for a quick risk dashboard.

```
from ttf_hh_spread import vol_sensitivity

vol_sensitivity(F_ttf=9.50, F_hh=9.00, T=0.25, r=0.045,
                sigma_ttf=0.60, sigma_hh=0.50, rho=0.35,
                option_type="call")
# {'base':        1.4129,
#  'σ_ttf +5%':   1.4744,
#  'σ_ttf −5%':   1.3553,
#  'σ_hh  +5%':   1.4563,
#  'σ_hh  −5%':   1.3751,
#  'ρ     +0.10': 1.3253,
#  'ρ     −0.10': 1.4942}
```

4.7 Display helper

`print_summary(res: SpreadResult) -> None`

Pretty-prints a full pricing report (underlyings, vols, premiums in USD/MMBtu **and** EUR/MWh, all Greeks, and Vega in the "per 1% vol move" market convention).

```
from ttf_hh_spread import spread_price, print_summary

res = spread_price(30.0, 3.0, 1.08,
                   T=180/365, r_usd=0.045,
                   sigma_ttf=0.60, sigma_hh=0.50, rho=0.35,
                   option_type="call")
print_summary(res)
# ================================================================
#   TTF / Henry Hub Spread Option — CALL
# ================================================================
#   ...
#   Option premium   :   6.3563 USD/MMBtu
#   Option premium   :  20.0821 EUR/MWh
#   ...
```

4.8 Put-call parity

For Margrabe exchange options, parity reads:

```
C − P = e^(-rT) · (F_TTF − F_HH)
```

Verification on the demo case (`F_TTF = 9.4955` , `F_HH = 3.0` , `T = 180/365` , `r = 0.045` , `σ_TTF = 0.60` , `σ_HH = 0.50` , `ρ = 0.35`):

Side	Value (USD/MMBtu)
`C − P`	6.352943
`e^(-rT)·(F_TTF − F_HH)`	6.352943

PCP holds to better than `1e-8 USD/MMBtu` .

---

5. `dashboard_jupyter.ipynb`

Interactive **Jupyter dashboard** built on top of the four pricing modules. Each section is a standalone widget panel: tweak the sliders/dropdowns and the prices, Greeks and charts re-render automatically.

5.0 Launching the notebook

```
# from the project root
pip install -r requirements.txt   # numpy, scipy, matplotlib, pandas,
                                  # ipywidgets, jupyterlab
jupyter lab dashboard_jupyter.ipynb
```

Then run the cells from top to bottom (`Shift + Enter`). Dependencies pulled in by the notebook:

Module	Used by
`black76_ttf`	Section 1 (Pricer), Section 5 (expiries)
`structures_ttf`	Section 2 (multi-leg structures)
`ttf_hh_spread`	Section 4 (Margrabe spread)
`ipywidgets`, `matplotlib`, `pandas`, `numpy`	UI + plots + tables

> ***Tip**: each section reads only its own widgets — you can edit / collapse any one of them without breaking the others.*

5.1 Section 1 — Pricer (Black-76 / Bachelier)

Single-leg European call/put pricer with full Greeks.

Inputs

Widget	Description	Default
`Model`	`Black-76` (lognormal) or `Bachelier` (normal)	Black-76
`Forward (EUR/MWh)`	Underlying TTF forward `F`	35.00
`Strike (EUR/MWh)`	Exercise price `K`	35.00
`Maturity (years)`	`T`, ACT/365	0.25
`Rate (decimal)`	Risk-free rate `r` (annualised)	0.03
`Vol (lognormal)`	Black-76 σ (visible only when `Model = Black-76`)	0.50
`Vol (EUR/MWh)`	Bachelier σ_n (visible only when `Model = Bachelier`)	10.00

Outputs

- Call and put premiums in EUR/MWh.
- Full Greek block for both call and put: `delta`, `gamma`, `vega`, `theta` (per calendar day), `rho` (per percentage point), `vanna`, `volga`.

Behaviour

- The vol slider auto-toggles between lognormal (Black-76) and normal (Bachelier) when the model dropdown changes — only the relevant one is shown.
- Recomputation is triggered on slider release (`continuous_update=False`), so the panel stays responsive on slow links.

- Internally calls `b76_call`, `b76_put`, `b76_greeks` (resp. `bach_*`) from `black76_ttf.py`.

5.2 Section 2 — Structures (10 multi-leg payoffs)

Pricer + P&L chart for the standard option structures wrapped in `structures_ttf.py`. Black-76 only, fixed `r = 3 %`.

Available structures

#	Structure	Strike inputs
1	Straddle	`K`
2	Strangle	`K_put`, `K_call`
3	Bull Call Spread	`K_lo`, `K_hi`
4	Bear Put Spread	`K_lo`, `K_hi`
5	Butterfly	`K_lo`, `K_mid`, `K_hi`
6	Condor	`K1`, `K2`, `K3`, `K4`
7	Collar	`K_put`, `K_call`
8	Risk Reversal	`K_put`, `K_call`
9	Calendar Spread	`K` (`T_far = T + 3 months`)
10	Ratio Spread 1×2	`K_lo`, `K_hi`

Inputs

- `Structure` dropdown — picking a new structure resets the strike sliders to sensible defaults around the forward (e.g. `F − 4, F, F + 4` for a butterfly).
- `Forward (EUR/MWh)`, `Maturity (years)`, `Vol (lognormal)`.
- 1 to 4 strike sliders `K1 … K4` whose **labels and visibility adapt** to the selected structure.

Outputs

- **Net premium** in EUR/MWh, labelled `debit` if positive, `credit` if negative.
- **Net Greeks**: `delta`, `gamma`, `vega`, `theta` (per calendar day).
- **Breakevens** (forward levels at expiry where P&L = 0).
- **Max profit** / **max loss** (`+inf` / `−inf` for unbounded structures).
- **P&L chart at expiry** with:
- solid blue line = total payoff − net premium,
- dashed grey line = current forward,
- dotted red lines = breakevens,

- light green vertical lines = strikes,
- green/red shading = profit/loss zones.

> ***Tip**: Calendar Spread is the only structure that uses two maturities — the slider drives `T_near`, while `T_far = T_near + 0.25 yr` is fixed inside the dispatcher.*

5.3 Section 3 — Vol Surface (parametric, 3D)

Parametric implied volatility surface, displayed as an interactive 3D `matplotlib` plot.

Model

ATM term structure plus a smile in log-moneyness:

```
atm(T)     = sigma_inf + delta_sigma · exp(−kappa · T)
m(K, T)    = ln(K / F) / sqrt(T)
sigma(K,T) = atm(T) + skew · m + wings · m²        (clipped to [0.01, 5.00])
```

The forward is held constant across the whole grid (a true forward curve could be plugged in via `ttf_market_data` later).

Inputs

Widget	Meaning	Range
`Forward (EUR/MWh)`	`F` (constant across the grid)	5 – 100
`sigma_inf (long ATM)`	long-end ATM vol	0.10 – 1.00
`delta_sigma (short ATM)`	additional short-end ATM bump	0.00 – 1.00
`kappa (decay)`	exponential decay rate of the bump	0.1 – 8.0
`skew`	linear smile slope vs log-moneyness	−0.40 – +0.40
`wings (convexity)`	quadratic smile coefficient	0.00 – 0.50
`range strikes (± %)`	strike grid half-width vs `F`	10 % – 90 %
`T max (years)`	longest maturity on the surface	0.25 – 5.0

Outputs

- A 3D surface plot (axes: `K`, `T`, `σ`) with the `viridis` colormap and a colourbar.
- Two info lines below the chart: `ATM(T=0)`, `ATM(T=∞)`, the achieved vol range and the grid size (default 35 strikes × 25 maturities).

> ***Tip**: a negative `skew` (default `-0.08`) reproduces the classic TTF shape — OTM puts trade richer than OTM calls. Increase `wings` to fatten both tails.*

5.4 Section 4 — TTF / Henry Hub Spread (Margrabe)

UI wrapper around `ttf_hh_spread.spread_price` (see Part 4). Prices and plots both the **payoff at expiry** and the **current option value** along the TTF axis, with a secondary x-axis in EUR/MWh.

Inputs

Widget	Description	Default
`Option`	`call` or `put`	call
`Forward TTF (EUR/MWh)`	TTF forward in EUR/MWh	35.00
`Forward HH (USD/MMBtu)`	Henry Hub forward in USD/MMBtu	3.50
`Vol TTF (lognormal)`	σ_TTF	0.50
`Vol HH (lognormal)`	σ_HH	0.45
`Correlation rho`	TTF/HH correlation ρ	0.50
`Maturity (years)`	`T`, ACT/365	0.50
`FX EUR/USD`	spot FX used to convert TTF to USD/MMBtu	1.080
`Rate USD`	USD risk-free rate	0.04
`Market price (USD/MMBtu)`	optional input for implied correlation	0.00

Outputs

- TTF forward shown in **both** EUR/MWh and USD/MMBtu, plus the spread vol `σ_spread = √(σ_TTF² + σ_HH² − 2·ρ·σ_TTF·σ_HH)`.
- Premium in **both** USD/MMBtu and EUR/MWh.
- All Margrabe Greeks: `delta_ttf`, `delta_hh`, `gamma_ttf`, `vega_ttf`, `vega_hh`, `vega_rho`, `theta` per calendar day.
- **Implied correlation**: if `Market price > 0`, the panel solves `implied_correlation` and prints the result. If the quote is outside the achievable corridor, an explanatory error is shown instead.
- A **payoff vs option-value chart**:
- blue solid line = payoff at expiry vs `F_TTF` (USD/MMBtu),
- orange dashed line = current Margrabe value at the chosen `T`,
- vertical reference lines at `F_TTF` (grey) and `F_HH` (red),

- top x-axis = `F_TTF` translated back into EUR/MWh.

5.5 Section 5 — Expiry Dates (ICE TFO)

Calendar viewer for the official ICE TFO option expiries (product code **TFO**).

Rule: `expiry = (1st of delivery month) − 5 calendar days`; if that day is a non-business day or non-UK business day, roll back to the nearest prior UK business day; if equal to the futures LTD, roll back one extra UK business day.

Inputs

Widget	Description	Default
`Number of contracts`	how many forthcoming expiries to display	12
`Reference`	reference date for the lookup	today

Outputs

A `pandas` DataFrame with one row per contract:

Column	Meaning
`Contract`	ICE code (`TTFK26` , `TTFM26` , …)
`Delivery`	delivery month (`May-26` , `Jun-26` , …)
`Option expiry`	ISO date of the option expiry
`Days to expiry`	`(expiry − reference) + 1` calendar days
`T (years)`	`T = Days / 365`
`Futures LTD`	last trading day of the underlying future

A **horizontal bar chart** displays `T (years)` for each contract, annotated with the calendar-day count.

> ***Backed by***: *`ttf_next_expiries`, `ttf_time_to_expiry`, `_ttf_futures_ltd` from `black76_ttf.py`.*

---

6. Financial glossary

Every market, model, Greek, structure and numerical convention used in the project, grouped by topic. Cross-references in *italics* point to other glossary entries.

6.1 Markets and underlyings

- **TTF** (*Title Transfer Facility*) — Dutch virtual natural gas hub operated by Gasunie Transport Services. The European benchmark for natural gas.
- **Henry Hub (HH)** — Physical natural gas pipeline interconnection in Erath, Louisiana. The US benchmark, settlement reference for NYMEX natural gas futures.
- **ICE Endex** — Exchange listing the *TTF* futures (product code TFM) and the **TTF Options** product (product code **TFO**), whose official expiry rule uses a UK business-day calendar.
- **Forward** — Agreed price today for delivery on a future date. Notation `F`. Implicitly already includes carry and interest, hence Black-76 prices options on `F` rather than on a spot price.
- **Future / Futures contract** — Exchange-listed, marked-to-market version of a forward. One TTF futures contract = physical delivery over a given delivery month.
- **Delivery month** — The calendar month over which a TTF futures contract delivers gas (e.g. `Jun26` → June 2026).
- **LTD** (*Last Trading Day*) — The last business day on which the future itself can be traded. The TTF option expires **before** the futures LTD.
- **LNG** (*Liquefied Natural Gas*) — Cross-basin transportable form of natural gas. LNG flows between TTF and HH drive the spread vol via the *netback* relationship.
- **Netback** — Arbitrage-style equality between two regional prices net of liquefaction, shipping and regasification costs. Drives convergence of TTF and HH and therefore the *implied correlation* on the spread.

6.2 Option mechanics

- **Option** — Right (without the obligation) to transact an underlying at a fixed price.
- **Call** — Option to **buy** the underlying at strike `K`. Payoff at expiry: `max(F_T − K, 0)`.
- **Put** — Option to **sell** the underlying at strike `K`. Payoff at expiry: `max(K − F_T, 0)`.
- **European style** — Exercise allowed **only at expiry**. The whole library assumes European options.
- **American style** — Exercise allowed at any time up to expiry (not used here).
- **Strike** (`K`) — Pre-agreed exercise price.
- **Maturity / Expiry** (`T`) — Date / remaining time at which the option can be exercised. Expressed in years (*ACT/365*).
- **Premium** — Price paid for the option (in EUR/MWh or USD/MMBtu).
- **Debit / Credit** — A multi-leg structure with a *positive* net premium (cost) is a *debit*; with a *negative* net premium (proceeds) it is a *credit*.
- **Intrinsic value** — Immediate exercise value: `max(F − K, 0)` (call) or `max(K − F, 0)` (put). What an option would be worth if expiry were now.

- **Time value** — `Premium − Intrinsic`. Positive while *T > 0*.
- **ATM** (*At-The-Money*) — `K = F`.
- **ITM** (*In-The-Money*) — Call: `K < F`. Put: `K > F`. Positive intrinsic.
- **OTM** (*Out-of-The-Money*) — Call: `K > F`. Put: `K < F`. Zero intrinsic.
- **Moneyness** — Generic ratio describing the relationship between `K` and `F`. *Log-moneyness* `m = ln(K / F) / √T` is used by the vol smile.
- **Payoff** — Cashflow at expiry as a function of the terminal forward `F_T`.
- **P&L at expiry** — Payoff *net of the premium paid*, used in *Section 2* of the dashboard.
- **Breakeven** — Forward level at expiry at which net P&L = 0.
- **Put-call parity (PCP)** — Model-free identity: `C − P = e^(−rT) · (F − K)`. Holds for both Black-76 and Bachelier; the Margrabe analogue is `C − P = e^(−rT) · (F_TTF − F_HH)`.

6.3 Pricing models

- **Black-Scholes (1973)** — Foundational lognormal model for stock options with continuous dividends.
- **Black-76 (1976)** — Adaptation of Black-Scholes for options on **forwards** (no spot, no carry): $C = e^{(-rT)}\cdot[F\cdot N(d_1) - K\cdot N(d_2)]$ with $d_1 = [\ln(F/K) + ½\sigma^2 T] / (\sigma\sqrt{T})$, $d_2 = d_1 - \sigma\sqrt{T}$. Default model in this project for $F \gg 0$.
- **Bachelier (1900)** — Normal (arithmetic) Brownian-motion model: $dF = \sigma_n \cdot dW$. Allows **negative forwards**, hence the right model for TTF in crisis or for spreads.
- **Margrabe (1978)** — Closed-form pricer for an *exchange option* on two log-normal forwards. See *Spread vol*.
- **Risk-neutral measure** — Probability measure under which discounted forwards are martingales. All pricers in the library evaluate expectations under that measure.
- **Discount factor (DF)** — `e^(−rT)`. Multiplies the forward expectation to convert it into a present value.
- **Risk-free rate** (`r`) — Annualised continuously compounded rate. Decimal (`0.02 = 2 %`).
- **No-arbitrage bounds** — A call price must lie in `[max(0, e^(−rT)(F−K)), e^(−rT)·F]`. The implied-vol solvers raise `ValueError` if a quote breaches them.
- **N(·) / Φ(·)** — Standard-normal CDF.
- **φ(·) / n(·)** — Standard-normal PDF.

6.4 The Greeks

Sensitivities of the option price `V` with respect to its inputs.

- **Delta (Δ)** — `∂V / ∂F`. Hedge ratio in forward units. Discounted by `e^(−rT)` here. `Δ_call − Δ_put = e^(−rT)`.
- **Gamma (Γ)** — `∂²V / ∂F²`. Convexity in the forward; identical for call and put. Peaks ATM and decays toward expiry.
- **Vega (ν)** — `∂V / ∂σ`. Sensitivity to volatility. Reported here per unit (1.00 = 100 %) of vol; divide by 100 for the market "per 1 vol point" convention.
- **Theta (Θ)** — `∂V / ∂t`. Time decay. Reported **per calendar day**. Negative for a long option.
- **Rho (ρ)** — `∂V / ∂r`. Sensitivity to the risk-free rate. Reported per *percentage point* of rate change.
- **Vanna** — `∂²V / (∂F ∂σ) = ∂Δ / ∂σ`. Cross-derivative used in skew-aware risk management. Zero ATM under Bachelier.
- **Volga** (also *Vomma*) — `∂²V / ∂σ²`. Vol convexity, drives the smile-roll risk.
- **Charm** / **Color** / **Speed** — Higher-order Greeks not used here but occasionally referenced in the literature.
- **Net Greeks** — Greeks aggregated across the legs of a multi-leg structure (Section 2 of the dashboard).

6.5 Volatility

- **Volatility (σ)** — Standard deviation of returns of the underlying.
- **Lognormal vol** — σ used in *Black-76*; dimensionless, decimal (`0.50 = 50 %`).
- **Normal vol** (σ_n) — σ used in *Bachelier*; absolute units, EUR/MWh. Rule of thumb: `σₙ ≈ F · σ` for matching ATM prices.
- **Implied volatility (IV)** — Vol level that, plugged into the model, reproduces a quoted market price. Solved by *Brent's method* in the library (`b76_implied_vol`, `bach_implied_vol`).
- **ATM volatility** — IV at `K = F`.
- **Smile** — Convex shape of `IV(K)` as a function of strike (or log-moneyness).
- **Skew** — Slope of the smile. The TTF smile is typically negatively skewed (OTM puts trade richer than OTM calls).
- **Wings** — Quadratic / convex contribution to the smile, fattening the tails on both sides.
- **Term structure** — Variation of ATM IV with maturity `T`.
- **Vol surface** — Two-dimensional `IV(K, T)`. Section 3 of the dashboard builds a parametric one: `σ(K,T) = sigma_inf + Δσ·exp(−κT) + skew·m + wings·m²`.

- **25-delta / 50-delta / 75-delta strike** — Strike whose call delta equals the given target. Standard quote convention; obtained via `b76_delta_to_strike` / `bach_delta_to_strike`.
- **Risk reversal** — Difference of IVs between the 25Δ call and the 25Δ put; measures *skew*. Also the name of a multi-leg structure (see 6.7).

6.6 Spread and correlation

- **Spread** — Difference of two related prices. Here `F_TTF − F_HH` after unit conversion to USD/MMBtu.
- **Spread option** — Option whose payoff depends on a spread, e.g. `max(F_TTF − F_HH, 0)`.
- **Exchange option** — Option to swap one asset for another at expiry, i.e. `max(F₁ − F₂, 0)`. Margrabe (1978) gives a closed form.
- **Margrabe formula** — Closed-form price of an exchange option: `C = e^(−rT)·[F₁·N(d₁) − F₂·N(d₂)]` with `d₁ = [ln(F₁/F₂) + ½σₛ²T] / (σₛ√T)`, `d₂ = d₁ − σₛ√T`.
- **Spread vol (σ_s)** — Effective lognormal vol of the spread: `σₛ = √(σ₁² + σ₂² − 2·ρ·σ₁·σ₂)`. Used inside Margrabe.
- **Correlation (ρ)** — Instantaneous correlation between the two log-forwards. $\rho \to +1$ → assets co-move → spread vol shrinks → option cheaper.
- **Implied correlation** — ρ backed out of a market spread-option price via `implied_correlation` (Brent on a monotone function of ρ).
- **MWh / MMBtu** — Energy units. `1 MWh = 3.412142 MMBtu`. TTF quotes in EUR/MWh, HH in USD/MMBtu, hence the conversion in `ttf_eur_to_usd` / `ttf_usd_to_eur`.
- **FX EUR/USD** — Exchange rate (e.g. `1.08`). Combined with the energy conversion to compare TTF and HH on the same axis.

6.7 Multi-leg structures

Eleven structures bundled by `structures_ttf.py` and exposed in Section 2 of the dashboard.

- **Straddle** — Long call + long put at the **same** strike `K`. Pure vol bet, max profit unbounded, max loss = total premium.
- **Strangle** — Long OTM put `K_put` + long OTM call `K_call`. Cheaper than a straddle, requires a wider move to break even.
- **Bull Call Spread** — Long call `K_lo` + short call `K_hi`. Bounded bullish payoff for a *debit*.
- **Bear Put Spread** — Long put `K_hi` + short put `K_lo`. Bounded bearish payoff for a *debit*.

- **Butterfly** — Long `K_lo` + short 2 × `K_mid` + long `K_hi` (all calls or all puts). Pin-risk play, max profit at `K_mid`.
- **Condor** — Four-strike generalisation of the butterfly: long `K1` + short `K2` + short `K3` + long `K4`. Wide pin zone.
- **Collar** — Long put `K_put` + short call `K_call`. Caps both upside and downside, near-zero premium for a hedger of the underlying.
- **Risk Reversal** — Short put `K_put` + long call `K_call`. Synthetic long forward; price reflects the *skew*.
- **Calendar Spread** — Short near-dated option + long far-dated option at the same strike. Sells short-dated theta against long-dated vega.
- **Ratio Spread (1×2)** — Long 1 × `K_lo` + short 2 × `K_hi`. Cheap or even *credit* directional bet with naked-leg risk past `K_hi`.
- **Net premium** — Sum of premiums across the legs (sign-aware).
- **Breakevens / Max profit / Max loss** — Reported by the dashboard for each structure (`+inf` / `-inf` for unbounded structures).

6.8 Conventions and numerics

- **Calendar / 365** — Time `T` to expiry is computed as `(expiry_date - reference).days / 365` — calendar days only, no business-day adjustment.
- **Business day** — Mon–Fri excluding **UK public holidays only** (England & Wales). Implemented in `ttf_is_business_day` : New Year's Day, Good Friday, Easter Monday, the early-May, spring and summer bank holidays, Christmas Day and Boxing Day (with the standard substitution to the next weekday when a fixed-date holiday falls on a weekend).
- **ICE month codes** — `F G H J K M N Q U V X Z` for January–December. Combined with a 2-digit year: `TTFM26` = June 2026.
- **Reference date** — Date from which `T` is measured. Defaults to `date.today()` in the helpers.
- **Brent's method** — Bracketing root-finder used for *implied volatility* and *implied correlation* (`xtol = 1e-8` , max 300 iterations).
- **Finite-difference Theta** — Theta is computed by `V(T - 1/365) - V(T)` rather than analytically, to stay consistent across the four pricers.
- **Sign convention** — Greeks reported with their natural sign: long call $\Delta > 0$, long option $\Theta < 0$, etc.
- **Numerical accuracy** — All round-trip identities (PCP, IV → price → IV, spread PCP) are verified to better than `1e-8` in the built-in tests.

Appendix A — Reference values (3 months, ATM)

With `F = 30 EUR/MWh, K = 30, T = 0.25, r = 0.02`:

Model	sigma	Call	Put	Δ_call	Γ	ν
B76	0.50	2.9670	2.9670	+0.547	0.0525	5.907
Bach	15 EUR/MWh	2.9770	2.9770	+0.498	0.0530	0.199

Theoretical `C − P` = `e^(-0.02·0.25)·(30 − 30) = 0`. Verified on both models (difference < 1e-12 EUR/MWh).

Appendix B — Reference values for the TTF/HH spread (Margrabe)

With `F_ttf_eur = 30 EUR/MWh`, `F_hh = 3 USD/MMBtu`, `fx = 1.08`, `σ_TTF = 0.60`, `σ_HH = 0.50`, `ρ = 0.35`, `r_usd = 0.045`, `T = 180 / 365`:

Quantity	Value	Unit
`F_ttf_usd`	9.4955	USD/MMBtu
Spread `F_TTF − F_HH`	+6.4955	USD/MMBtu
`σ_spread`	63.25%	—
Call premium	6.3563	USD/MMBtu
Call premium	20.0821	EUR/MWh
Put premium	0.0034	USD/MMBtu
`Δ_TTF` (call)	+0.9757	—
`Δ_HH` (call)	−0.9694	—

PCP check: `C − P = 6.352943 = e^(-rT)·(F_TTF − F_HH)`, verified to `< 1e-8 USD/MMBtu`.